# PG No. 36 $D_{3d}(1)$ $\bar{3}m$ (-31m setting) [ trigonal ]

* character table

| $D_{3d}(1)$ | $1(1)$ | $2_{120}(3)$ | $3^+_{001}(2)$ | $-1(1)$ | $m_{120}(3)$ | $-3^+_{001}(2)$ |
|---|---|---|---|---|---|---|
| $A_{1g}$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $A_{2g}$ | 1 | −1 | 1 | 1 | −1 | 1 |
| $E_g$ | 2 | 0 | −1 | 2 | 0 | −1 |
| $A_{1u}$ | 1 | 1 | 1 | −1 | −1 | −1 |
| $A_{2u}$ | 1 | −1 | 1 | −1 | 1 | −1 |
| $E_u$ | 2 | 0 | −1 | −2 | 0 | 1 |

* polar ↔ axial conversion

$A_{1g}$ ($A_{1u}$) $A_{2g}$ ($A_{2u}$) $E_g$ ($E_u$) $A_{1u}$ ($A_{1g}$) $A_{2u}$ ($A_{2g}$) $E_u$ ($E_g$)

* symmetric product

| | $A_{1g}$ | $A_{2g}$ | $E_g$ | $A_{1u}$ | $A_{2u}$ | $E_u$ |
|---|---|---|---|---|---|---|
| $A_{1g}$ | $A_{1g}$ | $A_{2g}$ | $E_g$ | $A_{1u}$ | $A_{2u}$ | $E_u$ |
| $A_{2g}$ | | $A_{1g}$ | $E_g$ | $A_{2u}$ | $A_{1u}$ | $E_u$ |
| $E_g$ | | | $A_{1g}+E_g$ | $E_u$ | $E_u$ | $A_{1u}+A_{2u}+E_u$ |
| $A_{1u}$ | | | | $A_{1g}$ | $A_{2g}$ | $E_g$ |
| $A_{2u}$ | | | | | $A_{1g}$ | $E_g$ |
| $E_u$ | | | | | | $A_{1g}+E_g$ |

* anti-symmetric product

| $A_{1g}$ | $A_{2g}$ | $E_g$ | $A_{1u}$ | $A_{2u}$ | $E_u$ |
|---|---|---|---|---|---|
| – | – | $A_{2g}$ | – | – | $A_{2g}$ |